



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212) Unit 1 –
Fundamentals of OOP
Topic – 1.4 C++ Programming Basics

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What is C++?

- C++ is a compiled, object-oriented language
- It is the "successor" to C,
 - (the "++" is called the successor operator in C++)
- C was derived from a language called B which was in turn derived from BCPL
- C was developed in the 1970's by Dennis Ritchie of AT&T Bell Labs
- C++ was developed in the early 1980's by Bjarne Stroustrup of AT&T Bell Labs.
- Most of C is a subset of C++



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

- *Compiler directive:*
Tells the compiler what to do before compiling
- This one includes source code from another file



Simple C++ Program

- *Main function*

```
#include <iostream>

int main()
{
    // Declarations
    // Statements
    return 0;
}
```



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

- Header for main function
- States...
 - data type for the return value
 - identifier for function
 - list of arguments between parenthesis (none for this function)



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

- Braces enclose the body of the function
- They represent the start and end of the function



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

- Declarations and statements
- Main body of function
- “//” represents the start of a comment



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

Return statement:

- specifies the value the function returns
- All (almost) declarations and statements end with a semi-colon ";"



Simple C++ Program

```
#include <iostream>
int main()
{
    // Declarations
    // Statements
    return 0;
}
```

- This program doesn't do anything!



Sample C++ Program

```
#include <iostream>
int main()
{
    int number;
    cout << "Enter a number" <<endl;
    cin >> number;
    cout << "You entered: " <<number;
    return 0;
}
```

Variable declaration:

- The identifier `number` is declared as being of data type `int`



Sample C++ Program

```
#include <iostream>

int main()
{
    int number;
    cout << "Enter a number" <<endl;
    cin >> number;
    cout << "You entered: " <<number;
    return 0;
}
```

- cout:
the output statement for C++
- Note the direction of "<<"
- endl:
represents an end-of-line



Sample C++ Program

```
#include <iostream>

int main()
{
    int number;
    cout << "Enter a number" << endl;
    cin >> number;
    cout << "You entered: " << number;
    return 0;
}
```

- cin:
the input statement
for C++
- Note the direction
of ">>"



Sample C++ Program

```
#include <iostream>

int main()
{
    int number;
    cout << "Enter a number" <<endl;
    cin >> number;
    cout << "You entered: " <<number;
    return 0;
}
```

Output:

Enter a number:

10

You entered:10

Thank You..